Course Code	21CSC555J	Course Name	MACHINE LEARNING ALGORITHMS	Course Category	C	PROFESSIONAL CORE	L	T	P	C
							3	0	2	4

Pre-requisite Courses	*Nil*	Co-requisite Courses	*Nil*	Progressive Courses	*Nil*
Course Offering Department	*Data Science and Business Systems*	Data Book / Codes / Standards	*Nil*		

Course Learning Rationale (CLR):	***The purpose of learning this course is to:***
CLR-1:	*Understand the Human learning aspects.*
CLR-2:	*Acquaintance with primitives in the learning process by computer.*
CLR-3:	*Develop the linear learning models and classification in machine learning*
CLR-4:	*Implement the clustering techniques and their utilization in machine learning*
CLR-5:	*Implement the tree-based machine learning techniques and to appreciate their capability*

Course Outcomes (CO):	***At the end of this course, learners will be able to:***	***Programme Outcomes (PO)***		
		1	***2***	***3***
CO-1:	*Demonstrate knowledge of learning algorithms and concept learning through implementation for sustainable solutions of applications.*	1	2	2
CO-2:	*Evaluation of different algorithms on well formulated problems along with stating Valid conclusions that the evaluation supports.*	2	3	3
CO-3:	*Formulate a given problem within the Bayesian learning framework with focus on Building lifelong learning ability.*	3	3	3
CO-4:	*Analyze research-based problems using Machine learning techniques and Apply different clustering algorithms used in machine learning to generic datasets and Specific multidisciplinary domains.*	1	3	2
CO-5:	*Apply decision tree learning algorithms.*	2	3	3

Module-1 - Introduction And Types of Learning	***15 hour***
Introduction: Machine Learning: What & Why? - Examples of Machine Learning applications, Training versus Testing, Positive and Negative Class, Cross-validation. Types of Learning: Supervised, Unsupervised and Semi-Supervised Learning. The Curse of dimensionality -Over fitting and under fitting-Linear regression-Bias and Variance tradeoff-Regularization-Learning-Evaluation Metrics, Gradient Descent Curve-Classification-Error and noise-Parametric vs. non-parametric models-Linear Algebra for machine learning. *Lab 1: Building programs to work with the data pre-processing in python* *Lab 2: Building programs to work with linear regression in python* *Lab 3: Building programs to work with cross validation in Python*	
Module-2 - Design And Analysis of Machine Learning Algorithms	***15 Hour***
Guidelines for machine learning experiments, Cross Validation (CV) and resampling – K-fold CV, measuring classifier performance, assessing a single classification algorithm and comparing two classification algorithms - K-fold CV Performance metrics-MSE, accuracy, confusion matrix, precision, recall, F1-score- Linear Regression - Logistic Regression with Binomial & Multi class classification *Lab 4: Building programs to performance metrics in python* *Lab 5: Building programs work with linear regression with multiple variables in Python* *Lab 6: Building programs work with logistic regression in python*	
Module-3 - Non-Parametric Model	***15 Hour***
K nearest neighbor classification –Gaussian Naive Bayes Classification-Multinomial Naïve Bayes classification-Bernoulli Naïve Bayes Classification - Comparison of Gaussian, Multinomial, Bernoulli naive	

bayes classification -Support vector machine.

Lab 7: Building python programs to use principal component analysis
Lab 8: Building python programs to use Naïve Bayes classification
Lab 9: Building programs to use Support Vector Machine

Module-4 – Decision Tree & Ensembling models **15 Hour**

Decision tree representation-Basic decision tree learning algorithm-Inductive bias in decision tree Decision tree construction-Issues in decision tree-Classification and regression trees (CART)- Random Forest-Random Forest with scikit-learn Minority Class, bootstrapping, Impurity Measures – Gini Index and Entropy, Best Split -Multivariate adaptive regression trees (MART). Boosting Technique- Working of Adaboost, Gradient boost, Extreme Gradient Boost, Stacking and voting ensemble models

Lab 10: Building programs to implement decision tree algorithm
Lab 11: Building programs to implement random forest algorithm and hyper-tunning the parameters
Lab 12: Building programs with Boosting models

Module-5 - Unsupervised Techniques **15 Hour**

Measuring (dis)similarity-Evaluating output of clustering methods-Spectral Clustering-Hierarchical Clustering-Agglomerative clustering-Divisive clustering-Choosing the number of clusters-Clustering data points and features-Bi-clustering-Multi-view clustering-K-Means clustering-K-medoids clustering, Principal component Analysis, Singular value decomposition (SVD).

Lab 13: Building programs to implement Hierarchical clustering
Lab 14: Building programs to implement K-Means clustering
Lab 15: Building program to implement PCA, SVD.

Learning Resources

1. *Géron, "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow", O'Reilly Media, Third Edition, 2022.*
2. *Ethem Alpaydin, "Introduction to Machine Learning", MIT Press, Fourth Edition, 2020.*
3. *Tom Mitchell, "Machine Learning", McGraw-Hill, 1997. Sebastian Raschka, Vahid Mirji, Python Machine Learning and deep learnin, 2nd edition, kindle book, 2018*
3. *Carol Quadros,"Machine Learning with python, scikit-learn and Tensorflow", Packet Publishing, 2018.*
4. Gavin Hackeling,"Machine Learning with scikit-learnll, Packet publishing", O'Reily, 2018.
5. *Stephen Marsland, "Machine Learning: An Algorithmic Perspective, "Second Edition", CRC Press, 2014.*
6. *Kevin P. Murphy," Machine Learning: A Probabilistic Perspective", MIT Press, 2012.*
7. *https://github.com/ageron/handson-ml3*
8. *https://www.kaggle.com/learn/intro-to-machine-learning*
9. *https://docdrop.org/download_annotation_doc/AAAMLP-569to.pdf*
10. *http://14.139.161.31/OddSem-0822-1122/Hands-On_Machine_Learning_with_Scikit-Learn-Keras-and-TensorFlow-2nd-Edition-Aurelien-Geron.pdf*

Learning Assessment

	Bloom's Level of Thinking	Continuous Learning Assessment (CLA) Formative CLA-1 Average of unit test (45%)		Continuous Learning Assessment (CLA) Life-Long Learning CLA-2 (15%)		Summative Final Examination (40% weightage)	
		Theory	Practice	Theory	Practice	Theory	Practice
Level 1	Remember	20%	-	-	20%	20%	-
Level 2	Understand	20%	-	-	20%	20%	-
Level 3	Apply	30%	-	-	30%	30%	-
Level 4	Analyze	30%	-	-	30%	30%	-
Level 5	Evaluate	-	-	-	-	-	-
Level 6	Create	-	-	-	-	-	-
	Total	100 %		100 %		100 %	